

	B[K, N]												b478 3790 3dc0 3d68 392a bc7f 3760 bd13 b6ee b081 b687 359e	3b99 3045 b90e bc13 b779 bb10 bcfc bc3d ba28 b8a0 b393 3c2a
													bde2 3449 3249 3db3 bb8e 39f2 bddb b45a 3dbd b566 b507 35f7	b943 2137 b99d 2cdd bd5f 3cb9 aab4 bd96 3a2b 3594 3024 383b
													3c0e 3b57 b7b3 bcd2 3c48 3b68 b87d 3d8c 31c0 bcb9 b8ea 3bca	28b0 3c28 32b9 b83e bcab 35bd 2cfd aef6 b6c0 baf8 bd1e 354a
A[M, K]	C[M, N] += A[i, k] @ B[k, j]													
3cf4 b6de 33e5 b700 b871 b597 ba92 b22b 314d 3800 b9a8 bc30 ba35 bb2f 393d 36c3 b943 a8fd bbd2 3986 2701 bdce 3cf4 aff8	3cd2 3dc7 bc9c 37a9 30ee b887 34b8 3d9a bddf 39f2 3be5 3a04 3dab bca2 31a2 3507 bdbb 3a67 3cc8 3a1a 3add bd6f 3920 38bc 320a b7d3 3b0e 30dc 3c48 bd99 bd9c b90e b863 3b0b 339b 3c40 ba4c 340c 3df2 3751 39db bc5a 3c16 3b15 3c1d 3da9 b4b8 373e bb69 3c09 b283 3da9 ba6b 3525 34ab bdd9 bc45 37d9 bd31 3d2a													
3a06 3c1f 35ae b39d b51d 3cf7 3183 bb17 3899 b902 b5b2 bc10 347c b9a3 3b23 3d8d 3b9f 3633 b762 3c1c 369c 38b2 bc89 b548	bd49 3c9c 3b4e b7d7 b8f6 3b19 3db5 35a9 b99d 38e7 399e ba49 b89c 3c0d 381b b6ea 2a20 399c bb1a bcd6 b0d9 bcd5 bc1e b847 3add 371a 390b 3db8 b13c 3aa4 ba46 b136 3814 341f bc15 bc4d b213 3b01 bd05 3914 bcc1 bdb8 3709 bcc8 3b72 b9c5 bbbd 3416 385e 3bbb b328 aef6 3d56 b981 bd10 bb59 3d5c 3d4e 3d65 bc08													
2f42 bd13 b45a bc8d bcd9 3d2d b49f bc94 b96b 3828 39f4 3d93 3c78 3058 bb5c 38ee 2d03 397f bb3d bac7 363a b023 a388 bcf8	38ec 3803 bdd1 b643 bb3f 39f9 ba06 b199 3923 b99c bc99 b92f 3d99 39b1 3369 ba49 3c0a bcb8 3b09 3cec b827 3d78 3578 392c b84f b1e7 ba06 b9ee 3dfd bc79 38c8 3884 3c83 b6fd 2b33 bc48 b9b4 b572 3ddb bbad b817 ba39 3dae 36b4 bc58 22e3 bb16 39d0 b97f b9c2 3961 b8fe bd28 b565 bc2a 3cd8 3a17 bdb6 3948 b1da													
3d30 b725 3a28 2ed8 bc61 bd1c 3406 b8a3 39c9 3c4a 3d8b 3c42 38dd 351b b087 3911 b925 bddf 39ff 2fe9 3913 3d43 3910 3ccc	33ab b0b2 bd7c bc2e 37cc ba9c 3dc5 b525 b625 38d6 b3f3 b4c7 3d74 bde8 bdb0 b5bb b9d3 b869 bdf0 3660 3b15 3722 ba03 bc7b b5e7 33cd 330d 359d 36a4 abd3 bb2f b161 b407 3c60 adc2 bd06 3a76 3cb4 b9cd 39a7 bcd1 3cc2 3a92 b86f 3d63 b66a bb9f 2f8a 365f b7ba 384e 3416 bc7c 381a afd5 389f b6b1 b79d bb82 b7a6													
b542 3746 bbf8 bc08 b582 b6c8 3d99 b2da bc39 b8a8 3c64 3809 3c7a 3c3f 3bc5 bd01 bbd3 bd45 3d85 3d21 3545 b735 a951 bcb7	3369 bb4a 3df2 3c05 bae1 3ce0 399b 35d6 bdac 2652 3997 3d6b bc99 3444 b8b2 39da 9efb 396f ba52 3ad3 39b6 3b9d 3974 359e bc74 3875 3ac4 a21f 3abe 3c9e bb91 bcc9 3dce bb73 bc38 38a4 bd38 b790 35ce b52c 32aa b65a b7b1 3b68 3d9e 3949 b545 b544 baf9 b77c 3d89 3baf 3cf0 39f5 bdc6 35dd 3493 b2bf 2ef3 3bb4													
3887 ad8b bbc2 355a b850 3cb5 b1bd 386c 3cd2 3da9 3b68 b123 b67a bcc2 b508 2d87 3874 3a79 34df 3d36 b474 32bc 276f bc3b	b51b 3d40 3d5d bd16 b3af bab6 b73c 2c48 bcfd 383f fffffb4c2 3736 b82c bd37 32c9 bd82 3c2a 2b35 b0df 2f7d bb7b 3d0b 3b0c 3c19 3169 3d14 bc1a bc86 bdaa b8be ba59 b9be bdd8 3d4a 2be5 3c4a b050 b9d2 b446 bbcd 3cf1 3a7a adf1 bd98 b6f0 37ca b4e2 34ba b6ee b5f7 baf0 3cca 3c9c 3c68 bbae 3899 3d0a b9ba b830													

```
acc += tl.dot(a_tile, b_tile) # fp16 tiles, fp32 accumulator, one C tile per program
```

Kernel Engineering

From silicon to speculative decoding – GPU kernels for modern LLMs. □